DS	**Session**	Searching	**Question Information**	Level 1 Challenge 4

Question Description:

Ragu has given a range [L, R] to Smith. Smith wants to require to find the number of integers 'X' in the range such that GCD(X, F(X)) > 1 where F(x) is equal to the sum of digits of 'X' in its hexadecimal (or base 16) representation.

Example : F(27) = 1+B=1+11=12

(27 in hexadecimal is written as 1B)

Constraints:

1 <= T <= 50

1 <= L

R <= 10^5

Input Format:

The first line contains a positive integer 'T' denoting the number of questions that you are asked.

Each of the next 'T' lines contain two integers 'L' and 'R' denoting the range of questions.

Output Format:

Print the output in a separate lines exactly 'T' numbers as the output.

Sample Input:

3

1 3

5 8

7 12

Sample output:

2

4

6

Explanation

For the first test-case, numbers which satisfy the criteria specified are : 2,3. Hence, the answer is 2.

For the second test-case, numbers which satisfy the criteria specified are : 5,6,7 and 8. Hence, the answer is 4.

For the third test-case, numbers which satisfy the criteria specified are : 7,8,9,10,11 and 12. Hence, the answer is 6.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
3
1 3
5 8
7 12
```

Test Case 2

INPUT (STDIN)

```
10
15 64
14 100
9 79
19 63
3 83
5 61
18 93
```

EXPECTED OUTPUT

```
2
4
6
```

```
20 98
13 71
12 53
```

EXPECTED OUTPUT

```
29
52
45
26
53
37
45
47
36
26
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
int search(int a, int b)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
2
```

Test Case 2

TOKEN COUNT

```
165
```

Test Case 3

NLOC

```
27
```